

A kinetic Monte Carlo method for the simulation of massive phase transformations

C. Bos^{*}, F. Sommer, E.J. Mittemeijer

Max Planck Institute for Metals Research, Heisenbergstrasse 3, D-70569 Stuttgart, Germany

Received 26 January 2004; received in revised form 22 March 2004; accepted 7 April 2004

Available online 6 May 2004

Abstract

A multi-lattice kinetic Monte Carlo method has been developed for the atomistic simulation of massive phase transformations. Beside sites on the crystal lattices of the parent and product phase, randomly placed sites are incorporated as possible positions. These random sites allow the atoms to take favourable intermediate positions, essential for a realistic description of transformation interfaces. The transformation from fcc to bcc starting from a flat interface with the fcc(1 1 1)//bcc(1 1 0) and fcc[1 1 1]//bcc[0 0 1] orientation in a single component system has been simulated. Growth occurs in two different modes depending on the chosen values of the bond energies. For larger fcc–bcc energy differences, continuous growth is observed with a rough transformation front. For smaller energy differences, plane-by-plane growth is observed. In this growth mode two-dimensional nucleation is required in the next fcc plane after completion of the transformation of the previous fcc plane.

© 2004 Acta Materialia Inc. Published by Elsevier Ltd. All rights reserved.

Keywords: Massive transformation; Monte Carlo techniques; Austenite–ferrite transformation; Transformation kinetics

1. Introduction

The massive transformation is a long-range diffusionless phase transformation during which the matrix phase is converted to a product phase with a different crystal structure but with the same composition [1]. This massive transformation is accomplished by diffusional jumps across the interface between the two phases. In many iron-based alloys the austenite, γ , to ferrite, α , transformation is of massive nature. Recently, the massive transformation has received much renewed attention [1–3]. The mechanism of the transformation is a topic of active discussion [4]. Most experimental data were collected after the transformation was completed. Data on the interface structure obtained during the transformation would greatly enhance the understanding of the transformation mechanism, but such data are scarce [4,5].

Molecular dynamics (MD) simulations would allow the study of the structure of a moving interface in full atomic detail, as simulations of the martensitic transformation have shown [6–8]. Unfortunately the timescale required for massive transformations is too demanding computationally. In MD simulations the full continuous trajectories of the atoms are described. To enhance the timescale that can be simulated, kinetic Monte Carlo (kMC) simulations can be made. However, then the possible atom positions are restricted to a set of discrete sites. The movement of the atoms is then treated as a stochastic process.

Monte Carlo methods have been employed successfully in simulating *solid–vapour* and *solid–liquid* transformations [9–13]. The simulations have provided detailed information on the interface structure during crystal growth for different types of materials [12,13]. Unfortunately these methods cannot be applied directly to *solid–solid* transformations. Because a kinetic Ising model provides the basis of kMC simulations it follows that the atoms are placed on a rigid lattice [14,15]. Hence, a description of a *solid–solid* transformation requires at least two lattices. Until now such an approach

^{*}Corresponding author. Tel.: +49-711-6893327; fax: +49-711-6893312.

E-mail address: c.bos@mf.mpg.de (C. Bos).

was computationally too demanding [16]. However, increased computational power makes it possible to develop a full multi-lattice method now and such a method is presented for the first time in this paper.

For a massive γ to α transformation simulation, two different crystal lattices are required: fcc and bcc. A crucial extra imposition is the use of random sites in the method proposed here. These sites are distributed randomly throughout the system. This allows the occurrence of deviations of the crystal structure arrangement, in particular at the transformation front (see Section 2).

This paper thus describes a multi-lattice kinetic Monte Carlo model for the simulation of a massive transformation (Section 2). To describe the atomic interactions a nearest neighbour bond counting model is used (Section 2.1). Although the model is not at all material specific, whenever materials properties had to be chosen, values that pertain to iron were selected. The first simulations performed with this model reported here were devised to test the applicability of this type of atomistic simulation.

The focus is on the transformation from the fcc to the bcc ($\gamma \rightarrow \alpha$) phase of a single element, starting with a flat, incoherent interface. Although a substantial amount of the bcc phase is present in the start configuration of the simulations, nucleation can still play an important role. A completely flat interface may require the formation of (two-dimensional) nuclei for further growth of the bcc phase. As described in Section 3, simulations have been performed to study such two-dimensional nucleation behaviour and to investigate the possibility of growth without such a separate nucleation phase. The results of the simulations are presented in Section 4. The discussion in Section 5 focuses on the nucleation and the growth behaviours.

2. Simulation method

As compared to earlier simulation models for crystal growth (e.g. see [10]) the introduction of intertwining multiple crystal lattices is the cardinal step to be made in order to simulate *solid–solid* phase transformations. As described below, not only multiple lattices but also randomly distributed sites have to be introduced as possible positions for the atoms.

A system is created with a partially filled fcc (γ) lattice and a partially filled bcc (α) lattice. This means that the initial nucleation of the bcc phase is not simulated (see Sections 1 and 3). A γ/α interface with a specific orientation is created. The events considered in the kMC model are atom jumps from a filled site to an empty site. The energy change ΔU associated with a jump is calculated on the basis of a nearest neighbour bond counting model as described in Section 2.1. The jump probability p is taken as [14]

$$p = \frac{\exp\left(-\frac{\Delta U}{k_B T}\right)}{1 + \exp\left(-\frac{\Delta U}{k_B T}\right)}, \quad (1)$$

where k_B is the Boltzmann constant and T is temperature. The simulation algorithm presented below has been developed from the following considerations.

Two sites (and atoms) are considered to be neighbours if the distance between them is less than $1.05r_f$ (the fcc nearest neighbour distance). In the simulations $r_f > r_b$ (r_b is the bcc nearest neighbour distance), the factor 1.05 has been chosen to prevent round-off errors. An atom is considered an interface atom if it has a neighbour of a type different from its own type or if it is next to a vacancy. Thereby the atoms next to vacancies are included in the collection of “interface” atoms in order to make movement of vacancies possible also when these vacancies are not positioned directly at the interface.

To prevent atoms from jumping through or into other atoms, not all empty neighbour sites can be valid target sites. There are two types of valid target sites. For the first type the site has to be within a radius r_{sj} of the source site. A jump to this type of target site is called a short jump. For the second type the target site has to be within the maximum jump distance $r_{maxjump}$ from the source site and there should be no other atom within a radius r_{vh} of the target site. If r_{vh} is chosen large enough then these target sites can be considered to be vacancies. These long jumps can then be considered to be vacancy jumps. The different ranges discussed are illustrated in Fig. 1. A jump from site A to site B is valid (short jump) as site B is within r_{sj} of site A. A jump to site C is not valid because site D is within r_{vh} of site C.

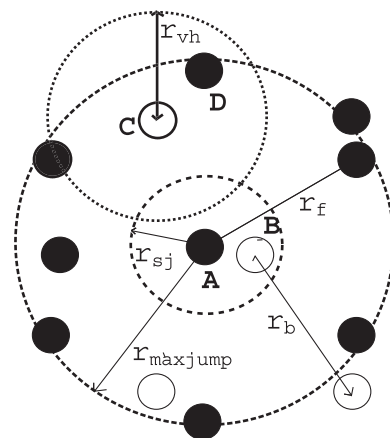


Fig. 1. The different ranges. Occupied fcc sites (black) and empty bcc sites (white) are shown. The maximum jump distance is $r_{maxjump}$ for a long and r_{sj} for short jump (shown for atom A), r_{vh} is the minimum radius required for a vacancy hole. The bcc and fcc nearest neighbours distances are r_b and r_f , respectively.

The distinction between these two types of valid jumps has been made for computational efficiency reasons. It is unlikely that the target site of a short jump will have an occupied neighbour within r_{vh} (as the jumping atom is so near). Therefore, the computationally costly procedure of checking for an occupied neighbour within r_{vh} is not performed for these target sites. To deal with the unlikely case that the target site of a short jump *does* have an occupied neighbour within r_{vh} the atoms have been given a hard-sphere core. When two atoms are within a distance smaller than the hard-sphere diameter d_{hs} , the atoms experience a (infinitely) large repulsion. This ensures that two atoms will never overlap (see also Section 2.1) because of a short jump.

All simulations have been done with $r_{sj} = 0.35r_b$, $r_{vh} = 0.85r_b$ and $r_{maxjump} = 1.025r_f$.

If only fcc and bcc lattice sites are included in the simulation then the transformation must be accomplished by direct jumps from fcc to bcc sites. However, recognizing the complexity of actual interface structures, it is considered possible that the atoms first visit an ‘intermediate’ position. The locations of these intermediate positions are not known and therefore difficult to include in the simulation. By including uniformly distributed randomly placed sites within the system it is assured that at least some of these extra positions are available for the atoms at the interface. The amount of available computer memory limits the number of random sites that can be included. Because these sites are distributed randomly they have a chance of being located at positions that will never be occupied. The efficiency of the random sites can be enhanced by continuously assigning new random positions until they have been occupied. In other words, the random sites ‘search’ for the (relevant) intermediate positions. Simulations have shown that without random sites no significant progress of the transformation is possible.

Finally, a time step is defined such that after one time step every interface atom has made one jump attempt.

The simulation algorithm then can be given as follows:

- (1) At random, select an interface atom.
- (2) At random, select a valid empty neighbour site of that interface atom.
- (3) Calculate the energy change ΔU for the jump to this empty site.
- (4) Calculate the corresponding jump probability p according to Eq. (1).
- (5) Generate a random number $r \in [0, 1)$ from a uniform distribution.
- (6) Accept the jump if $r < p$.
- (7) Continue step 1 until all interface atoms have been selected once and only once.
- (8) Update the time step counter by one.
- (9) Give the random sites that have remained unoccupied a new position.

- (10) Expand or contract the system in the direction perpendicular to the original interface (see Section 3).
- (11) Continue with step 1.

2.1. The atomic interaction model

The model for the atom interaction has been devised to comply with three requirements:

- (1) The bulk bcc atoms must have a lower energy than bulk fcc atoms.
- (2) It must cost energy to enlarge the interfacial area.
- (3) Atoms very close to each other must repel one another.

The atom–atom interactions are expressed in terms of bond energies u . As there are three different kind of sites, there are six different kinds of bonds. Three of these bonds have been given a fixed energy. These are the fcc–fcc bond-energy u_{ff} , the bcc–bcc bond-energy u_{bb} and the bcc–fcc bond-energy u_{bf} . Choosing specific values for u_{ff} and u_{bb} will fulfill requirement 1. For bcc–bcc and fcc–fcc the distance between two atoms is fixed (only nearest neighbours are considered), but for the fcc–bcc distance this is not the case. Yet, the fcc–bcc bond energy, u_{bf} , is taken as a constant. Requirement 2 can then be satisfied by choosing a specific value for u_{bf} (see also Section 3.1).

The three other kinds of bonds involve atoms on random sites (random–bcc, random–fcc and random–random). The interaction between a random site and a fcc/bcc/random site is taken independent of the distance between these two sites, but the value of the interaction is made variable as it depends on the surroundings of the random site(s) as follows. If a random site is very close to a specific lattice site, it should behave like that lattice site. Therefore, the energy of a bond with an atom on a random site depends on the distance of that random atom to the nearest lattice site, r_{ran} . The maximum distance between a random site and an xcc (where x is b or f) site is $\frac{1}{2}r_x$, the minimum distance is zero. At the maximum distance the bond-energy with an xcc atom is given by u_{rx} . For distances between the minimum and maximum values the bond energy for a random–xcc atom pair is interpolated linearly.

There are eight different interpolation equations. Four of these concern a bond between a lattice site of type x (b or f) and a random site that behaves like a site of type y (b or f)

$$u = u_{rx} + (\lambda u_{xy} - u_{rx}) \frac{\frac{1}{2}r_y - r_{ran}}{\frac{1}{2}r_y}, \quad (2)$$

where λ is a parameter that will be explained below. The other four concern bonds between a random site A that behaves like a site of type x, with a distance r_{ranA} to that site, and a random site B that behaves like a site of type y, with a distance r_{ranB} to that site

$$u = \frac{1}{2} \left[u_{rx} + (\lambda u_{xy} - u_{rx}) \frac{\frac{1}{2} r_y - r_{\text{ranB}}}{\frac{1}{2} r_y} + u_{ry} + (\lambda u_{xy} - u_{ry}) \frac{\frac{1}{2} r_x - r_{\text{ranA}}}{\frac{1}{2} r_x} \right]. \quad (3)$$

If the distance between a random site and the considered nearest lattice neighbour site, r_{ran} , is very small, the random site is almost indistinguishable from that lattice site; i.e. from an energy point of view it becomes indifferent if an atom occupies the lattice site or the random site. This means that the chance that such a random site becomes occupied is large. Often, it is necessary to count the number of atoms of a specific type: for example, the number of bcc atoms to determine the fraction transformed material. In this case the atoms on random sites very close to bcc sites should be counted as bcc atoms too. One option to ascertain that these occupied random sites close to the bcc lattice sites are recognized as bcc atoms, is to separately assign a crystal structure type to every single random atom based on the local surroundings of each atom. However, this not trivial task is computationally extensive. A second option is to ensure that the atoms have a significantly higher chance to occupy the lattice site as compared to the chance that the random site will be occupied. This can be accomplished with the parameter λ in Eqs. (2) and (3). By choosing a value smaller than one for λ , the maximum energy of a bond with an atom on a random site is reduced (less negative). In all simulations a value of 0.8 has been used for λ (i.e. for r_{ran} the bond-energy is 80% of u_{xy}).

Requirement 3 is fulfilled by giving the atoms a hard-sphere core. The hard-sphere diameter d_{hs} has been chosen as

$$d_{\text{hs}} = 0.68r_b. \quad (4)$$

With this value the hard-sphere core is just a little larger than the tetragonal interstitial holes in the bcc crystal structure. If two atoms are within the range d_{hs} of each other, they are given such a large (positive) bond-energy that a jump for which such a bond would be created is never accepted.

3. Simulation details

All interfaces have been created with the orientation fcc(111)/bcc(110) and fcc[111]/bcc[001]. The interface is completely flat and lies parallel to the xy -plane. A start configuration has three bcc planes and between 35 and 45 fcc planes. All simulations have been performed with imposed periodic boundary conditions in the x - and y -directions. The system volume is rectangular. The cross-sectional area of the xy -planes is kept constant

during the simulations; volume change is accommodated by expansion/contraction in the z -direction.

The bcc(110) and the fcc(111) lattice spacings ($d_{\text{b}(110)}$ and $d_{\text{f}(111)}$) are generally not equal. This means that as fcc planes are transformed to bcc the system wants to expand or contract (depending on the densities of the crystals) in the z -direction. The system is allowed to do this by shifting those fcc sites that have not yet made a jump trial in the simulation. This means that by this shift deviations from the ideal fcc lattice can occur because some atoms in one fcc plane are shifted while others are not. However, these deviations are so small that the crystal structure is not disturbed.

The shift of the fcc atoms is done once every five time steps, according to

$$\Delta z = \frac{N_{\text{newbcc}}}{N_{\text{bcc,plane}}} (d_{\text{b}(110)} - d_{\text{f}(111)}), \quad (5)$$

where N_{newbcc} is the number of newly formed bcc atoms and $N_{\text{bcc,plane}}$ is the number of atoms in a bcc plane. This procedure is consistent with having zero pressure acting on the system in the z -direction, implying that the system can freely expand or contract in this direction.

The fcc and bcc lattices have been created such that the numbers of atoms in an fcc(111) and a bcc(110) plane are equal. With the chosen number density of 0.103×10^{30} atoms/m³ (this is approximately 1.3 times the density of iron) for bcc and an initial interfacial area of 19.4×8.7 nm, the number of atoms per plane is 3312.

To avoid large heterogeneity of the distribution of the random sites over the entire system, the random sites are not distributed completely uniformly across the system. Instead a so-called restrictedly random distribution is adopted as follows. First, the system is divided in small cubic cells with a side length of $1.4r_b$. Then, 10 random sites are inserted into each cell according to a uniform distribution.

The simulation program was run on a single processor PC (Pentium 4 Xeon at 2.8 GHz).

3.1. Possible bond energy values

When the bond energies are chosen a number of conditions must be fulfilled. To ensure that bcc is energetically more favourable than fcc and recognizing the coordination numbers of the bcc and fcc lattices, the most important requirement is

$$8u_{\text{bb}} < 12u_{\text{ff}}. \quad (6)$$

Not every jump from an fcc to a bcc site leads to an energy decrease: this depends on the surroundings of the sites considered. An atom at the interface can have different numbers of bcc and fcc neighbours. When an atom only has a few bcc neighbours and a large number of fcc neighbours, the atom should favour an fcc site over a bcc site (and vice versa with many bcc neighbours).

In the simulation of crystal growth it is common to demand that an incoming atom (from the vapour or liquid) can only successfully attach on kink sites where it has a certain minimum number of solid neighbour atoms [10,13]. Here an analogous requirement is imposed: every atomic jump from fcc to bcc where an atom forms less than five bcc–bcc bonds should lead to an energy increase ($\Delta U_{\text{jump}} > 0$). With less than five bcc neighbour atoms the atom does not have enough ‘bcc character’ yet and the jump is unfavourable. If five or more bcc–bcc bonds are formed the jump is favourable and it is required that $\Delta U_{\text{jump}} < 0$. These conditions can be met by selecting proper values for u_{bb} , u_{ff} and u_{bf} . By looking at the number and type of the bonds that are broken and formed in a jump and in view of the requirement of inequality (6), ranges with valid values for u_{bb} , u_{ff} and u_{bf} can be indicated.

With the chosen interface orientation, there are for every short jump from an fcc to a bcc site where four bcc–bcc bonds are formed, seven fcc–fcc bonds broken (in case of a long jump a vacancy is involved and these numbers change). In case of five bcc–bcc bonds formed there are six broken fcc–fcc bonds (see Appendix A for an explanation of these numbers). The change in the number of mixed (fcc–bcc) bonds depends on the arrangement around the jumping atom. The energy change $\Delta U_{4\text{bb}}$ for a jump where four bcc–bcc bonds are formed is given by

$$\Delta U_{4\text{bb}} = 4u_{\text{bb}} - 7u_{\text{ff}} + \Delta N_{\text{mixed}}u_{\text{bf}}, \quad (7)$$

where ΔN_{mixed} is the change in the number of mixed fcc–bcc bonds. As it is demanded that for *every* short jump from fcc to bcc where four bcc–bcc bonds are formed $\Delta U_{\text{jump}} > 0$, only the limiting case of the smallest possible value for $\Delta U_{4\text{bb}}$ because of ΔN_{mixed} must be considered. All bond energies are negative, which makes $\Delta U_{4\text{bb}}$ minimal when ΔN_{mixed} is maximal. For the given interface orientation the maximum value for ΔN_{mixed} is four, which leads to

$$\Delta U_{4\text{bb},\text{min}} = 4u_{\text{bb}} - 7u_{\text{ff}} + 4u_{\text{bf}} \quad (8)$$

for the minimum energy change for a jump where four bcc–bcc bonds are formed. By introducing $x = u_{\text{bb}}/u_{\text{ff}}$ and using Eq. (8), the requirement $\Delta U_{4\text{bb},\text{min}} > 0$ can be written as

$$(4x - 7)u_{\text{ff}} > -4u_{\text{bf}}. \quad (9)$$

Because u_{bf} and u_{ff} are negative, it follows that $x < \frac{7}{4}$. Together with inequality (6) the range of valid values of x can be indicated by

$$\frac{3}{2} < x < \frac{7}{4}. \quad (10)$$

Hence, for a chosen value of u_{bb} , the range (10) prescribes the valid values for u_{ff} . Further rewriting inequality (9) as

Table 1

Bond-energy values in eV, u_{xy} is the energy of a bond between an atom on a site of type x and an atom on a site of type y (bcc, fcc or random)

u_{bb}	u_{ff}	u_{bf}	u_{rb}	u_{rf}
−1.04	−0.610	−0.0262	−0.0174	−0.0174

With the values given here, requirements (10) and (11) are fulfilled.

$$0 > u_{\text{bf}} > \left(\frac{4x - 7}{-4} \right) u_{\text{ff}} \quad (11)$$

the valid values for u_{bf} can be expressed as a function of x and u_{ff} .

For most simulations a value of −1.04 eV was selected for u_{bb} . The energy of a (bulk) bcc atom is then approximately equal to that of an iron bcc atom. Recognizing requirement (10), for x a value of 1.7 was selected, which corresponds with $u_{\text{ff}} = -0.610$ eV. Next, in view of requirement (11), u_{bf} was set to −0.0262 eV. The maximal (i.e. least negative) bond energies with random atoms, u_{rb} and u_{rf} , were set to values two-thirds of u_{bf} (see Table 1 for the complete set of values).

3.2. Nucleation

All simulations start with three planes of bcc atoms present, implying that three-dimensional nucleation is not required. However, because the start configuration is a flat interface, new bcc phase must nucleate in the fcc plane at the interface. This implies two-dimensional nucleation. The associated critical nucleus size has been determined by inserting more or less circular bcc seeds of different sizes into this fcc plane and measuring the energy change as a function of nucleus size. Then simulations of lateral growth have been performed. The simulations have been executed applying the bond-energy values from Table 1 and a temperature set to 1771 K.

3.3. Growth

The first atom in the fcc plane at the interface to jump to a bcc site only forms two bcc–bcc bonds (see Appendix A). Therefore, with the bond energies from Table 1, $\Delta U_{\text{jump}} > 0$ is positive and relatively large. This means that the chance that such a jump is accepted is very low ($\approx 10^{-16}$). Even for the next adjacent fcc atom in the interface plane considered that jumps to bcc and that thereby forms three bcc–bcc bonds, the chance for acceptance is still low ($\approx 10^{-9}$). Altogether, this makes the chance of two-dimensional nucleation very low and it would take years of calculation time to establish a substantial transformation. Therefore, simulations have been performed with the bond-energy values given in Table 2. With these values, jumps from fcc to bcc where three bcc–bcc bonds are formed have a negative ΔU_{jump} ,

Table 2

Bond-energy values in eV, u_{xy} is the energy of a bond between an atom on a site of type x and an atom on a site of type y (bcc, fcc or random)

u_{bb}	u_{ff}	u_{bf}	u_{rb}	u_{rf}
−0.959	−0.295	−0.0131	−0.00872	−0.00872

With these values inequality (10) is *not* fulfilled.

this violates inequality (10). To further examine the influence of the fcc–bcc energy difference, simulations have been performed for several (lower) values of u_{bb} , maintaining the other bond energy values as given in Table 2.

For all growth simulations the temperature was set to 1012 K and 10 vacancies were inserted randomly into the fcc plane at the interface. The vacancies were inserted because they have a positive influence on the growth rate as will become clear from the discussion of the results in Section 5.

4. Results

4.1. Nucleation

The calculated energy cost of inserting a more or less circular bcc nucleus of height of one atomic plane in the fcc plane at the interface is shown in Fig. 2 as a function of the nucleus radius. The size of the interface area parallel to the original interface does not change when a bcc nucleus is inserted in the fcc plane at the interface, also not upon its subsequent growth. Only the outer circumference of the nucleus contributes to a change in the total interface area. Therefore, the energy change for the creation of the circular nucleus, $\Delta U_{\text{nucleus}}$, can be written as

$$\Delta U_{\text{nucleus}} = \pi r^2 d \Delta U_V + 2\pi r d \gamma, \quad (12)$$

where r is the nucleus radius, ΔU_V is the fcc–bcc energy difference per unit volume, γ is the interface energy per unit area interface and d is the height of an atomic plane. A fit of Eq. (12) to the calculated data is shown too in Fig. 2 (dashed line). From the fit

$$\Delta U_V = (-7.4 \pm 0.4) \times 10^9 \text{ J/m}^3$$

and

$$\gamma = (4.5 \pm 0.2) \text{ J/m}^2$$

are obtained.

The simulations have demonstrated that nuclei with a radius smaller than the critical radius ($\approx 6.1 \text{ \AA}$) shrink and disappear and that nuclei larger than the critical size grow. It should be noted that this lateral growth is irregular (see Fig. 3). When vacancies are added in the neighbourhood of the nucleus the same kind of irregular growth is observed but the growth rate then is much higher (cf. Figs. 3 and 4). Growth is confined to the plane in which the nucleus was created and no other nuclei form spontaneously in the time of simulation.

4.2. Growth

Adopting the bond energies given in Table 2, the entire system quickly transforms continuously from fcc to bcc, as shown in Fig. 5. The insertion of a two-dimensional nucleus is not required; the bcc phase grows spontaneously. The transformation does not result in a perfect bcc crystal. A single bcc plane after transformation is shown in Fig. 6. The plane contains a few defects in the form of atoms placed on random sites, sometimes accompanied by one or two atoms on fcc sites.

When the fcc–bcc energy difference is lowered by giving u_{bb} less negative values, the transformation curve

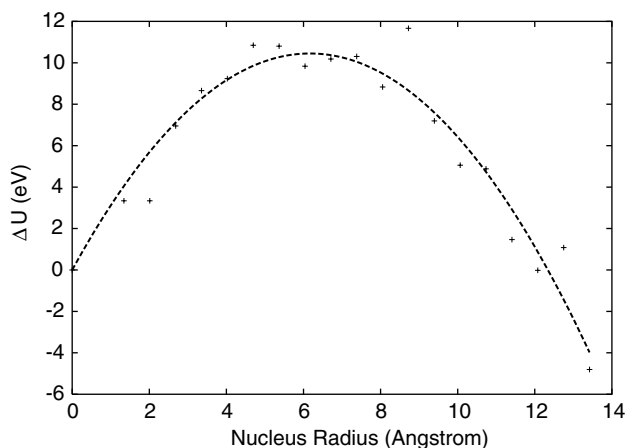


Fig. 2. Energy cost of building a bcc nucleus in an fcc plane at the bcc/fcc interface. Results of calculations with the bond energy values given in Table 1. The dashed line represents the (least squares) fit of Eq. (12) to the “measured” data points.

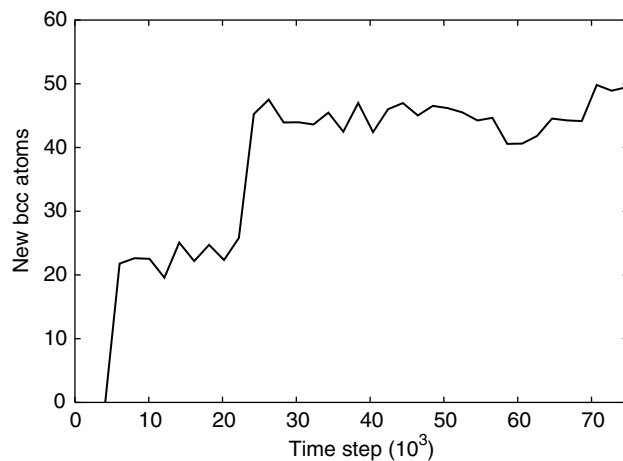


Fig. 3. Lateral growth of a bcc nucleus with a start radius of $10.7 \text{ \AA}$ (≈ 73 bcc atoms). Result of simulation with the bond energies from Table 1.

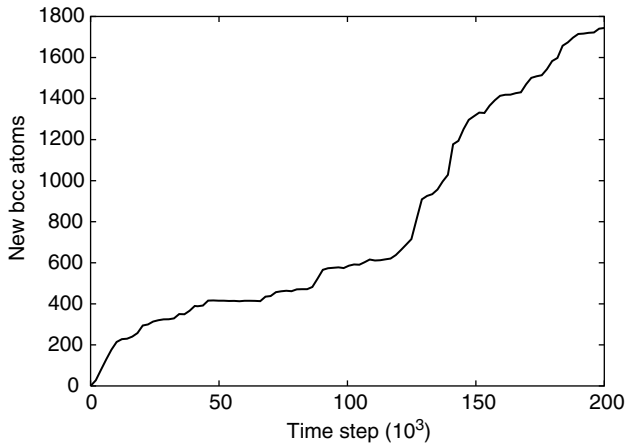


Fig. 4. Lateral growth of a bcc nucleus with a start radius of 10.7 Å ($\cong$ 73 bcc atoms), with seven vacancies in the vicinity of the nucleus. Result of simulation with the bond energies from Table 1.

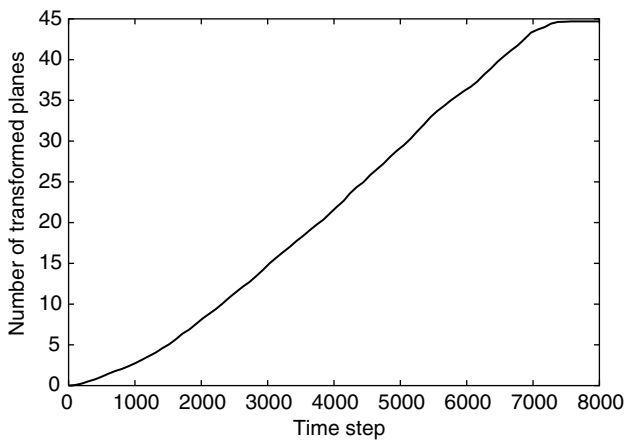


Fig. 5. Number of transformed planes as a function of time; result of simulation using the bond-energies from Table 2.

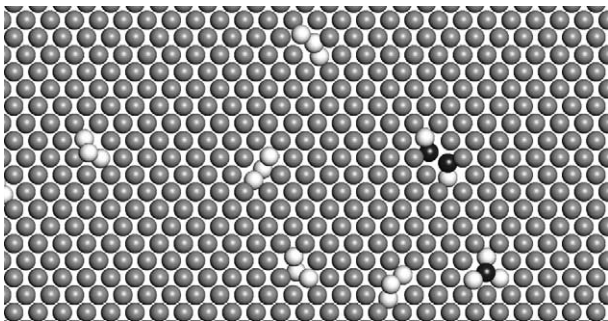


Fig. 6. A close-up of a single bcc plane, parallel to the x,y axes, after transformation (simulation with the bond-energies from Table 2). The black atoms are residual fcc atoms, the grey atoms are bcc atoms and the atoms on random sites are white. Atoms that seem to overlap are placed above each other (i.e. have different z coordinates).

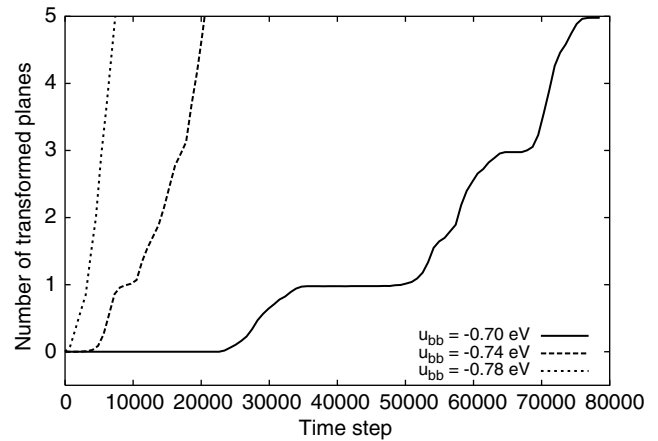


Fig. 7. Number of transformed planes as a function of time for three values of u_{bb} (the other bond energies are equal to the values given in Table 2).

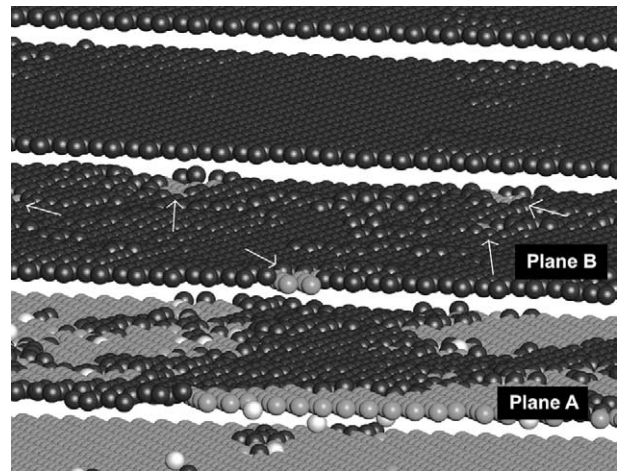


Fig. 8. A continuous fcc $\rightarrow$ bcc transformation in progress. The extra space between the planes is not real, it has been inserted to provide a clearer view. Fcc atoms are black, bcc grey and random atoms are white. The arrows indicate spots where the bcc phase is already growing in plane B before plane A has been transformed completely. Simulation with bond energies from Table 2.

changes significantly. For three different values of u_{bb} a part of the transformation curve is shown in Fig. 7. At $u_{bb} = -0.70$ eV the growth mode is no longer continuous, but is of plane-by-plane nature. In the continuous growth mode the transformation takes place in several planes simultaneously (Fig. 8). In the plane-by-plane growth mode, a single plane is completely transformed before bcc nucleates in the next fcc plane.

5. Discussion

5.1. Nucleation

The values that have been found for ΔU_V and γ can be directly related to the bond energy values (from

Table 1). Upon transformation the gain in bulk energy per unit volume of the nucleus considered is

$$\Delta U_V = N_{\text{transf}} \Delta U_{\text{bcc, fcc}} \quad (13)$$

with N_{transf} as the number of transformed atoms per unit volume of the nucleus and $\Delta U_{\text{bcc, fcc}}$ the (bulk) fcc–bcc energy difference per atom ($= \frac{1}{2}(8u_{\text{bb}} - 12u_{\text{ff}})$). From the values chosen for the bcc density and the bond energies (cf. Section 3 and Table 1) it thus follows:

$$\Delta U_V = -8.1 \times 10^9 \text{ J/m}^3.$$

This calculated, theoretical value for ΔU_V agrees fairly well with the value obtained from the fit of Eq. (12) to the simulated data: $(-7.4 \pm 0.4) \times 10^9 \text{ J/m}^3$. In the theoretical calculation a perfect circular disc is taken as the form of the nucleus. However, the real form of a nucleus can be different because it is made of a discrete number of atoms, which causes the difference between the measured (simulated) and theoretical values of the critical size.

For a theoretical prediction for the interface energy, the number of interface atoms per unit interface area, N_{iatoms} , must be known. To this end the numbers of fcc and bcc atoms adjacent to the nucleus–matrix interface have been counted (again only the interface area perpendicular to the outer circumference of the disc has to be considered cf. Section 4.1). For different nucleus sizes N_{iatoms} is not constant, but shows small deviations from the average value (in the range of nucleus sizes considered here) which is $3.45 \times 10^{19} \text{ atoms/m}^2$. N_{iatoms} consists out of two parts: fractions due to fcc atoms, f_{fcc} , and due to bcc atoms, f_{bcc} . These fractions are not constant but depend on the nucleus size. Therefore, average values will be used for f_{fcc} and f_{bcc} . Atoms at the disc circumference have, in comparison to the other atoms in the plane of the disc, depending on the local structure, one or two kin-like bonds replaced by mixed (fcc–bcc) bonds. Thus, as a first order approximation, it is assumed that for every atom at the disc circumference, upon becoming a disc circumference atom, one kin-like bond is replaced by a mixed bond, and γ can then be assessed as

$$\gamma = \left(\bar{f}_{\text{fcc}}(u_{\text{bf}} - u_{\text{ff}}) + \bar{f}_{\text{bcc}}(u_{\text{bf}} - u_{\text{bb}}) \right) \bar{N}_{\text{iatoms}} \quad (14)$$

with $\bar{f}_{\text{fcc}} = 0.58$, $\bar{f}_{\text{bcc}} = 0.42$, $\bar{N}_{\text{iatoms}} = 3.45 \times 10^{19} \text{ atoms/m}^2$ and the bond energies from Table 1, it then follows $\gamma = 4.2 \text{ J/m}^2$.

This value is close to the value obtained from the simulations $((4.5 \pm 0.2) \text{ J/m}^2)$.

The lateral growth of a two-dimensional nucleus is not smooth and continuous (see Figs. 3 and 4). This reflects that an increase in nucleus size beyond the critical size does not always lead to an energy gain, as demonstrated by the results shown in Fig. 2. Because of local variations

of the interface structure, the interface energy is not constant. Apparently, for certain nucleus sizes and shapes, the overall interface energy is so high that further growth again costs energy rather than delivers energy, although the nucleus size is already larger than the critical size. This effect can be ascribed to the limited number of different positions the atoms can take: the atoms can occupy random sites, but it is not guaranteed that these random sites are located at energetically favourable positions. Therefore, the interface is unlikely to be fully relaxed. This explanation is supported by the simulation with the vacancies added in the neighbourhood of the nucleus (cf. Fig. 4). The vacancies are dissolved quickly in the interface, thereby providing extra space at the interface and consequently more relaxation possibilities for the interface atoms. This explains the observed higher growth rate as compared to the simulation without vacancies (cf. Figs. 3 and 4).

5.2. Growth

Spontaneous growth of the bcc phase, within practical finite times of transformation, occurs on adopting the bond energy values from Table 2, as shown in Fig. 5. It is observed that during transformation of the first full five planes, the transformation rate increases gradually to a value that remains practically constant for the remaining part of the transformation. In the last part of the transformation the rate drops again because the total interface area starts to decrease as there is no more material to transform. Defects can occur in the transformed planes (Fig. 6), in particular for the first few transformed planes. Therefore, these planes contain a few more atoms than the normal total of 3312 (cf. Section 3). This corresponds with an increase of the number of vacancies in the untransformed part of the system. The presence of vacancies increases the growth rate (cf. discussion in Section 5.1), as is observed (Fig. 5).

Depending on the value of u_{bb} the growth mode is either continuous (see Fig. 7, $u_{\text{bb}} = -0.78 \text{ eV}$) or plane-by-plane (see Fig. 7, for $u_{\text{bb}} = -0.70 \text{ eV}$). In case of plane-by-plane growth it takes a certain time before a new bcc nucleus is formed in the next fcc plane adjacent to the transformation front. After nucleation, the fcc plane transforms relatively fast. In case of continuous growth, no such nucleation phase can be distinguished: the transformation front is a rough interface (see Fig. 8). Note that even in this case the product, fcc, phase is contiguous. At an intermediate value for u_{bb} (see Fig. 7, for $u_{\text{bb}} = -0.74 \text{ eV}$) the growth mode is mixed: for transformation of the first and second fcc planes a clear nucleation phase can be discerned, but then growth becomes continuous (the steps in the transformation curve disappear). Although not clearly shown in Fig. 7, the same effects happens for $u_{\text{bb}} = -0.70 \text{ eV}$ after longer simulation times. Equations similar to Eq. (8) can be

written for the energy changes ΔU_{2bb} and ΔU_{3bb} for jumps from an fcc site to bcc sites with two and three bcc neighbours, respectively

$$\Delta U_{2bb} = 2u_{bb} - 9u_{ff} + 8u_{bf} \quad (15)$$

and

$$\Delta U_{3bb} = 3u_{bb} - 8u_{ff} + 4u_{bf}. \quad (16)$$

Using Eqs. (15) and (16), the jump probabilities p_2 and p_3 for jumps to sites with two and three bcc neighbours, respectively, can be calculated by applying Eq. (1) as a function of u_{bb} ; the results are shown in Fig. 9. An fcc interface atom surrounded by fcc atoms only in its own (1 1 1) plane and that jumps to a bcc position makes a p_2 -jump (see also Appendix A). In the range -0.8 to -0.7 eV for u_{bb} , p_2 is very low (see Fig. 9) and hence only few of these jumps will be accepted. An accepted p_2 -jump is the first step of the formation of a new bcc nucleus. If p_3 is also very low then this one-atom-nucleus will quickly dissolve because no stabilizing extra bcc atoms will attach to it. The value for p_3 increases rapidly in the u_{bb} range from -0.8 to -0.7 eV and thus at $u_{bb} = -0.70$ eV the formation of a new bcc nucleus is much more difficult than at $u_{bb} = -0.78$ eV (cf. Fig. 7).

The change of growth mode upon continued transformation and the associated roughening of the transformation interface are consequences of the increase in the number of vacancies in the untransformed planes (see above discussion): the presence of vacancies makes nucleation easier, because energetically more favourable atom positions can be occupied (see also Section 5.1).

The surface roughening transition has been observed in crystal growth simulations with temperature as variable instead of a bond energy [9,11,13]. Obviously, according to Eq. (1), a change in p_2 and p_3 can be induced by a temperature change as well as by a change in u_{bb} . Hence a similar growth mode variation can be caused by both temperature and bond energy changes.

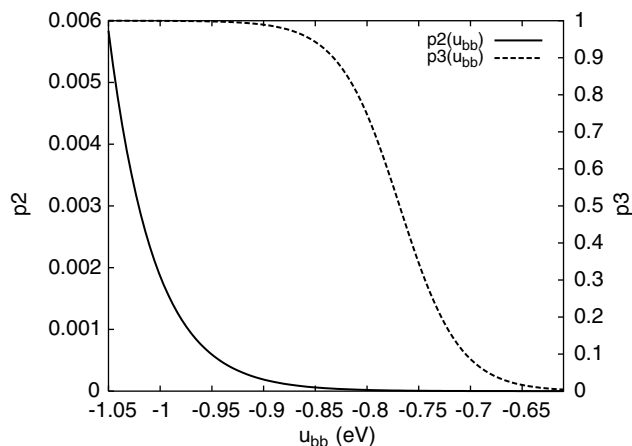


Fig. 9. The jump probabilities p_2 and p_3 as a function of u_{bb} (cf. Eq. (1) and Eqs. (15) and (16)), with other bond energies as indicated in Table 2 and at $T = 1012$ K.

5.3. Final remarks

The ledge mechanism has been suggested as a transformation mechanism for the massive transformation [1]. With ledges there is no need for two-dimensional nucleation as long as some mechanism operates that makes the formation or the preservation of ledges possible. The simulations with the laterally growing nucleus (Figs. 3 and 4) demonstrate that ledge-wise growth is a possible mechanism. Thus plane-by-plane growth can be conceived as a form of ledge-wise growth, in association with “spontaneous” formation of new ledges.

With the atomic interaction model used here (see Section 2.1) the energy of a bond between two atoms does not depend on the distance between those two atoms. As future work it is planned to use a more realistic atomic interaction model such as the Embedded Atom Method potential [17,18]. This potential is known to work well for metals, even for the description of defect structures. With the use of such a potential a molecular dynamics approach can then be used to examine separately the activation energy for different (individual) jumps, an aspect that has thus far not been taken into account.

6. Conclusion

For the first time a massive transformation has been simulated departing from a multi-lattice basis in a kinetic Monte Carlo approach. An essential ingredient of the model is the incorporation of “random sites”, not belonging to the product or parent lattices. In this way relevant intermediate atomic positions are identified and irregularities in the atomic structure of transformation interfaces can be accounted for.

Transformation mechanism and kinetics can be simulated for the massive fcc–bcc transformation adopting a bond counting model for the calculation of atomic jump probabilities. Starting with a completely flat initial interface, this model allows direct assessment of the critical nucleus size and interface energy values in terms of the bond energies.

Two different kinds of transformation mechanisms can be identified: plane-by-plane growth in association with two-dimensional nucleation, resembling the ledge mechanism, and continuous growth in association with the occurrence of a rough transformation interface. Which mechanism is dominant, depends on the values of the bond energies and the temperature. Increasing the fcc–bcc energy difference (i.e. making the energy difference more negative), causes the growth mode to change from plane-by-plane to continuous.

The transformation does not lead to a perfect bcc crystal. Point defects are formed especially in the initial stage of the transformation. This corresponds with an

increase of the vacancy concentration in the untransformed part of the system close to the interface. These vacancies influence the interface energy and lead to an increase of the transformation rate.

Appendix A. Calculation of the number of broken and formed bonds

In a bcc (110) plane an atom has four (nearest) neighbours. In the bcc crystal in the planes above and below this plane the atom has another two neighbours, leading to a total of eight neighbours. In an fcc (111) plane an atom has six neighbours and three in the fcc planes above and below, leading to a total of 12 neighbours. The planes near the interface have been labeled A, B, C and D. They are stacked in alphabetical order, A and B are bcc (110) planes and C and D are fcc (111) planes. The interface lies between plane B and C.

A bcc atom in plane B has four bcc neighbours in that plane and two bcc neighbours in plane A. The number of possible mixed fcc–bcc bonds with atoms in plane C turns out to be one, two or three depending on the local situation. For an fcc atom in plane C: six fcc–fcc bonds occur in plane C, three fcc–fcc bonds with atoms in plane D and one, two or three mixed fcc–bcc bonds with bcc plane B. The change in the number of bonds for a jump can be determined from these considerations. For example, when an fcc atom in plane C moves to a bcc site two new bcc–bcc bonds are formed (with atoms in plane B) and nine fcc–fcc bonds are broken (six with atoms in fcc plane C and three with atoms in fcc plane D). These broken fcc–fcc bonds are replaced with bcc–

fcc bonds, the exact number depends on the local arrangement.

Different example (cf. Section 3.1). Two bcc atoms in plane C, surrounded by fcc. If a common fcc neighbour (in plane C) also becomes bcc then there are four new bcc–bcc bonds formed (two with plane B and two with plane C) and seven fcc–fcc bonds broken (four with plane C and three with plane D). Mixed bonds still depend on the local arrangement.

References

- [1] Aaronson H. *Metall Mater Trans A* 2002;33:2285–97.
- [2] Massalski T. *Metall Mater Trans A* 2002;33:2277–83.
- [3] Liu Y, Sommer F, Mittemeijer E. *Acta Mater* 2003;51(2): 507–19.
- [4] Aaronson H, Vasudevan V. *Metall Mater Trans A* 2002;33:2445–70.
- [5] Howe J, Reynolds W, Vasudevan V. *Metall Mater Trans A* 2002;33:2391–411.
- [6] Entel P, Meyer R, Kadau K. *Philos Mag B* 2000;80(2):183–94.
- [7] Lill J, Broughton Q. *Phys Rev B* 2001;63:144102.
- [8] Kadau K, Germann T, Lomdahl P, Holian B. *Science* 2002;296:1681–4.
- [9] Leamy H, Jackson K. *J Appl Phys* 1971;42(5):2121–7.
- [10] Gilmer G, Bennema P. *J Appl Phys* 1972;43:1347–60.
- [11] Leamy H, Gilmer G. *J Cryst Growth* 1974;24/25:499–502.
- [12] Das A, Mittemeijer E. *Philos Mag A* 2001;81(11):2725–42.
- [13] Jackson K. *Interface Sci* 2002;10:159–69.
- [14] Müller-Krumbhaar H, Binder K. *J Stat Phys* 1973;8(1):1–24.
- [15] Binder K. *Monte Carlo Methods in Statistical Physics*. Berlin: Springer; 1986.
- [16] Levine S, Clancy P. *Modelling Simul Mater Sci Eng* 2000;8:751–62.
- [17] Daw M, Baskes M. *Phys Rev Lett* 1983;50:1285.
- [18] Daw M, Baskes M. *Phys Rev B* 1984;29:6443.